Gauss Te-Chuan Chang

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EDUCATION

B.S. in Physics & B.S. in Civil Engineering, National Taiwan University 2020-2025

- Dual bachelor's degrees in Physics and Civil Engineering, completing 205 credits.

M.S. in Structural Engineering, National Taiwan University 2025-2027

- Thesis: **Secondary Weakening in dm-Scale Laboratory Earthquakes: Frictional Origin or Rigid-Body Artifact?**

RESEARCH EXPERIENCES

Validation of Seismic Parameters via Laboratory Seismic Experiments 2024–Present
Division of Structural Engineering, Department of Civil Engineering, NTU Advisor: Dr. Chun-Yu Ke

- Performed all **finite-element simulations (ABAQUS)** of a double-direct-shear apparatus, showing that the acrylic loading block stores $> 10\times$ more strain energy than the granite blocks and dominates the energy release [EPSL, 2026].
- Led the **cohesive-zone-model** fits to near-fault strain records, showing near-front dissipation ($< 10\,\mathrm{J/m^2}$) more than an order of magnitude below the breakdown work.
- Designed and ran 1-D (50 cm) **PMMA laboratory-earthquake experiments** with strain-rosette arrays, and collaborated on 2-D (50×50 cm) experiments with piezoelectric (PZT) sensors; after an event at 28 MPa welded the fault, developed an optical laser-sheet method to detect frictional melting directly.
- Implemented a 3-D **dynamic rupture model** with rate-and-state friction in *tatva* (JAX, differentiable finite elements) verified on SCEC/USGS TPV101/102 against UGUCA and SPECFEM3D (∼1% RMS in slip).

Seismic Structural Health Monitoring and Analysis 2023–Present
Institute of Earth Sciences, Academia Sinica Advisors: Dr. Kuo-Fong Ma, Dr. Wen-Tzong Liang

- Contributed to **QSIS**, a low-cost MEMS seismic network in more than ten buildings in Taiwan, tracking building health with eigenfrequency, spectral, and deconvolution analyses.
- Identified a coherent 3.906 Hz **instrument artifact** below the noise floor by stacking cross-correlation functions; traced it to the data loggers' cooling fans, prompting a network-wide switch to passive cooling.
- Maintained seismic stations with IES technicians after the 2024 M_w 7.4 Hualien earthquake, co-locating QSIS sensors with observatory-grade seismometers.

AI-Based Seismic Signal Classification 2023–2025
Institute of Earth Sciences, Academia Sinica Advisors: Dr. Kuo-Fong Ma, Dr. Wen-Tzong Liang

- Trained CNN-based and a Transformer-based deep learning model to distinguish P-waves, S-waves, and non-seismic signals in real-time.
- Optimized the model for deployment on a Raspberry Pi 4B, enabling low-latency, server-independent earthquake detection on a cost-effective edge device.

N-Body Problem Simulation Research Autumn 2024
Department of Physics, NTU Advisor: Dr. Ying-Jer Kao

- Implemented a Barnes–Hut N-body simulation in Python, C++, and Fortran, validated through energy and angular-momentum conservation.

HONORS

Taiwan Farmers' Association Scholarship (four times) 2023–2026

Awarded to the top 300 students nationwide: as an undergraduate in 2023 and 2024, and as a graduate student in 2025 and 2026.

NTU Alumni Excellence Scholarship Summer 2024

Awarded for exceptional academic achievement at NTU, with a GPA of 4.23/4.3 in the second semester of 2022–23 (31 credits).

PUBLICATIONS

Peer-Reviewed Journal Publications

[1] Chun-Yu Ke, **Gauss T. Chang**, Gregory C. McLaskey, David S. Kammer, Chris Marone. (2026) Nonlocal Elastic Unloading as a Mechanism for Breakdown Work Scaling in Laboratory Earthquakes. ***Earth and Planetary Science Letters***, doi: 10.1016/j.epsl.2026.120283.

Manuscripts in Preparation

[1] **Gauss T. Chang**, Chun-Yu Ke. *Frictional melting and fault welding in decimeter-scale laboratory earthquakes on PMMA faults* (tentative title). In preparation.

[2] Wen-Tzong Liang, **Gauss T. Chang**, Utpal Kumar, Kuo-Fong Ma, et al. *QSIS: A low-cost structural health monitoring system* (tentative title). In preparation.

Selected Conference Abstracts

[1] **Gauss T. Chang**, Chun-Yu Ke. *Secondary Weakening in dm-Scale Laboratory Earthquakes: Frictional Origin or Rigid-Body Artifact?*. JpGU–AGU Joint Meeting 2026, Chiba, Japan, 24–29 May 2026.

[2] **Gauss T. Chang**, Chris Marone, Chun-Yu Ke. *Earthquake Stress-Drop Estimation in Laboratory Experiment with Machine Learning*. the Japan Geoscience Union Meeting 2025, Chiba, Japan, 25–30 May 2025.

[3] **Gauss T. Chang**, Wen-Tzong Liang, Utpal Kumar, Kuo-Fong Ma, Li-Wei Chen. *Orchestrating Structural Safety with QSIS: Vision for Extensible Building Arrays*. Japan Geoscience Union Meeting 2025, Chiba, Japan, 25–30 May 2025.

[4] Chun-Yu Ke, **Gauss T. Chang**, Gregory McLaskey, Chris Marone. *Earthquake Rupture Speed Dependence on Normal Stress in Laboratory Experiments*. EGU General Assembly 2025, Vienna, Austria, 27 April–2 May 2025.

[5] **Gauss T. Chang**, Wen-Tzong Liang, Utpal Kumar, Kuo-Fong Ma, Hung-Jun Chou, Li-Wei Chen. *QSIS: A Scalable and Cost-Effective Solution for Advanced Structural Seismic Monitoring*. AGU24 Annual Meeting, Washington, D.C., 9–13 December 2024.

[6] **Gauss T. Chang**, Chris Marone, Chun-Yu Ke. *Inferring Earthquake Stress Drops with Passive Acoustics in Laboratory Experiments Using Machine Learning*. Taiwan Geosciences Assembly 2025, Nangang, Taipei, Taiwan, 16–19 June 2025.

[7] **Gauss T. Chang**, Wen-Tzong Liang, Utpal Kumar, Kuo-Fong Ma, Hung-Jun Chou, Li-Wei Chen. *Natural Frequencies of Low-Rise Reinforced Concrete Buildings Determined With QSIS*. Taiwan Geosciences Assembly 2025, Nangang, Taiwan, 16–19 June 2025.

[8] **Gauss T. Chang**, Wen-Tzong Liang, Hung-Jun Chou, Utpal Kumar, Kuo-Fong Ma. *Toward Structural Health Monitoring with QSIS Network*. TEC24 Annual Meeting, Tainan, Taiwan, 23–24 October 2024.